

① Data Cleaning Sequence & Justification

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1. Cleaning Sequence

- * Standardize Date Formats: Convert mixed representations (DD/MM/YYYY) & MM-DD-YYYY into a single standard format.
- * Handle Missing Phone Numbers: Flag null values with a placeholder.

2. Justification of Sequence:

- * Standardizing dates first provides consistent formatting across records, enabling accuracy during the record matching step.

3. Impact of Skipping Steps:

- * Skipping Data: Causes software parsing failures, breaking chronological sorting & use.
- * Skipping Deduplication: Artificially inflates total customer counts, leading to inaccurate metrics and redundant marketing spend.

2) Outlier Analysis in Retail Sales

1. Assessment

The sale of ₹15,000,000 is an extreme outlier. It is 30 times larger than the standard upper range limit of ₹500,000.

2. Verification Methods

Interquartile Range (IQR): Compute $IQR = Q_3 - Q_1$,
Value $> Q_3 + 1.5 \times IQR$

Z-score Method: Calculate standard deviation.

$$Z = \frac{X - \mu}{\sigma}$$

Domain Audit: Check the raw logs to verify if the entry is a typographical error.

Treatment	Statistical Effect	Impact of Business Analysis
Keeping Raw Data	Inflates mean and standard deviation.	Distorts daily retail sales forecasts.
Direct Removal	Restores baseline statistical metrics.	Understands total revenue reconciliation

③ Strong Matching: Fuzzy Matching vs. RegEx

1. Direct Comparison

Feature	Regular Expressions ^(RegEx)	Fuzzy Matching
Mechanism	Deterministic syntax pattern matching.	Probabilistic distance scoring.
Output	Exact Binary Match	Proximity Score
Best	Standardized, rule-based formatting	Unstructured, typographic text variations.

2. Name Matching → Fuzzy Matching

Reason: Names contain missing letters variations, phonetic differences and short format.

Justification: Fuzzy algorithms calculate edit distances to similar names probabilistically.

2. Email Validating → RegEx Matching

Reason: Email addresses must adhere strictly to predefined internet syntax standards.

Justification: RegEx evaluates precise structural criteria. Fuzzy matching cannot determine if string syntax is structurally valid.

④ Correlation Analysis: Advertising vs. Sales

1. Dataset Exploration & Visualizations

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

2. Interpretation Business Action.

Strong

($r = 0.8$ to 1.0)



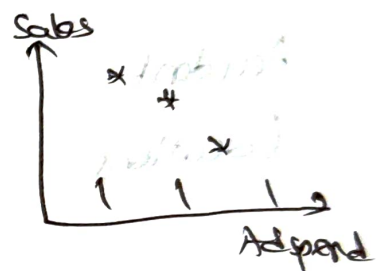
Weak

($r = 0.2$ to 0.8)



Negative

($r = -0.8$ to -1.0)



Strong positive: Ad spend reliably drive sales

~~growth~~

Growth.

Weak

Ad spend has little effects on sales.

Strong Negative: High spend aligns with lower

3 region sales.

5) Urban Planning Data Visualizations

Visualization Matrix & Decision Value

Urban Data Type	Recommended Visualization	Analytical Justification & Decision Impact
Monthly Traffic Volume	Line Chart	Tracks continuous trends and seasonality across time to low-volume months.
Accident Locations	Spatial Heatmap	Displays point density on a GIS map to pinpoint high-risk areas.
Population Distribution	Choropleth Map	Color-codes geographic zones to balance public transit, utility capacity & emergency.
Public Feedback Comments	Word Cloud Sentiment Bar Chart	Summarizes unstructured text into major topics ("potholes", "delays") & qualifies overall public satisfaction.